A Reproducible Methodological Framework for Prosecutorial Congestion Risk Prediction Using Explainable Machine Learning and Temporal Validation

Moisés Evangelista Gamarra[1*] and Jerremi Aron Chancan Labajos[1]

[1*]Servicio Nacional de Adiestramiento en Trabajo Industrial (SENATI), Lima, Peru.

*Corresponding author(s). E-mail(s): mevangelistag@senati.pe; Contributing authors: 152048@senati.pe;

Abstract

Congestion in justice systems constrains institutional planning because it distorts operational workload and the allocation of scarce resources. This study proposes and audits a reproducible machine learning framework that estimates the proxy risk of prosecutorial congestion from the administrative records of the Public Prosecutor's Office of Peru for the 2019–2026 period, comprising 9,593 annual operational units. The framework couples consensus-based feature selection over six complementary selectors with an explicit temporal availability audit, a year-block validation scheme, and a joint assessment of discrimination, calibration and decision-analytic utility. Every predictor is classified by the moment at which it becomes observable, and the twelve variables unavailable when the prediction must be issued are removed before any model is fitted. Each data partition carries a single documented role: training on 2019–2023, decision-threshold selection on 2024, one locked evaluation on 2025, and an exploratory check on the partial 2026 file. The selected XGBoost configuration attains a ROC-AUC of 0.796 (95% CI 0.760–0.838) and an F1-score of 0.409 (0.331–0.487) on the locked evaluation. Calibration analysis shows systematic over-estimation of risk above a predicted probability of 0.3, and decision curve analysis shows a negative net benefit at the F1-optimal operating point. The framework is therefore reported as a ranking and triage instrument rather than an alerting system.

Keywords: machine learning, prosecutorial analytics, judicial analytics, congestion risk, data leakage, model calibration, decision curve analysis, explainable artificial intelligence

1 Introduction

Congestion in justice systems represents a persistent challenge for institutional management: it lengthens response times, hinders the efficient allocation of resources, and constrains the operational capacity of the organisations responsible for administering justice. Administrative registries released under open data mandates now make this imbalance measurable at national scale, and machine learning offers a route to anticipating it. The question this study addresses is narrower and more demanding than it first appears: whether the imbalance of a coming year can be estimated from what is already known when that year begins.

Several studies have applied machine learning to predict judicial congestion, workload and institutional performance. Most, however, evaluate predictive algorithms, feature selection techniques or interpretability methods in isolation. Temporal validation, explicit control of data leakage, the joint assessment of robustness, and — most consequentially — the question of whether the resulting scores would support an actual decision, remain uncommon within a single reproducible workflow. The result is a body of reported performance figures whose operational meaning is hard to establish, because the reader cannot determine which information the model held at the moment the prediction is nominally issued, nor whether the predicted probabilities mean what their scale implies.

In response, this study proposes a reproducible Consensus, Temporal Validation and Explainability Methodological Framework (CTVEMF) to estimate the proxy risk of prosecutorial congestion in the Public Prosecutor's Office of Peru, and subjects it to three audits that are usually left implicit.

The first asks *what the model knows and when it knows it.* We define an explicit prediction time point and classify every candidate predictor as observable before, during or after the reporting year closes, admitting only the first class. This converts leakage control from an assurance into an auditable table.

The second asks *which data set played which role.* Each partition is assigned one documented function, so that no data set is used both to tune a decision and to certify it.

The third asks *whether the resulting score is useful, and at what price.* Discrimination, calibration and decision-analytic utility are reported separately, including the direction and not merely the magnitude of calibration error.

The contributions of this work are therefore: (i) a consensus-based robust feature selection strategy combining six methods, preceded by an explicit multicollinearity audit and confined to the earliest training block; (ii) a temporal availability audit that makes the information set of the model explicit at the level of individual variables; (iii) a year-block temporal validation scheme in which threshold selection and final evaluation are provably disjoint; (iv) a joint robustness assessment through an ablation study, stress testing and unsupervised validation of the feature space; and (v) a SHAP-based interpretability analysis reported together with a decision curve analysis that leads us to moderate, rather than assert, the operational value of the model.

The remainder of this paper is organised as follows. Section 2 presents the related work. Section 3 describes the methodology. Section 4 presents the results. Sections 5–7 present the interpretability and impact analysis, the limitations, and the conclusions.

2 Related Work

2.1 Machine Learning for Predicting Judicial and Prosecutorial Workload

Previous research on predicting case duration and procedural delay has focused predominantly on the judicial domain, largely overlooking prosecutorial dynamics. Recent studies show that machine learning can contribute to modelling operational workload and optimising institutional performance from administrative records, including work aimed at reducing judicial congestion, evaluating productivity and conducting temporal analyses of judicial processes [1–3]. Some combine tree-based models with explainability techniques to improve the interpretation of predictions; others incorporate process mining and temporal analysis to identify bottlenecks and measure procedural duration [1, 3]. Both lines report promising results, yet neither simultaneously integrates temporal validation, leakage prevention and robust feature selection within one methodological framework. Review work documents a broader transition from isolated predictive models towards comprehensive frameworks for the intelligent management of judicial systems [4].

Two limitations motivate the present study. The first concerns scope: the prosecutorial stage that precedes judicial proceedings receives markedly less attention than the judicial stage, even though it governs the volume that reaches the courts. The second concerns protocol: reported evaluations rarely state which information was available at the moment the prediction is nominally issued, which makes it difficult to judge whether the reported figures would survive deployment.

2.2 Feature Selection: Filter, Wrapper, Embedded, and Ensemble Approaches

Feature selection occupies a central role in building robust predictive models because of its effect on stability, interpretability and generalisation. Filter methods score predictors independently of any learner and are inexpensive but blind to interactions. Wrapper methods exploit the predictive performance of a specific classifier at substantially greater computational cost [5]. Embedded methods balance the two by selecting during training, though their behaviour continues to depend on the algorithm employed. Because individual selectors disagree in the presence of correlated variables, and because that disagreement yields unstable subsets, consensus and ensemble strategies have been proposed to aggregate several criteria into a more stable decision [5, 6]. Among all-relevant embedded selectors, the Boruta algorithm compares the importance of each feature against randomised shadow copies within a Random Forest, providing a statistically supported relevance test rather than a minimum-optimal subset [7].

The proposed framework implements a consensus over six complementary methods — two filters (mutual information and ANOVA), one filter for categorical variables (χ^2), one embedded method (Random Forest) and two wrappers (Boruta and RFECV) — preceded by an explicit multicollinearity audit using the Pearson correlation coefficient and Cramér's V. What this work adds is a constraint on *when* selection may inspect the data: the selectors are fitted only on the earliest training block and checked on a later year that remains internal to the training partition.

2.3 Class Imbalance, Data Leakage, and Temporal Validation

Risk classification on administrative data typically exhibits substantial class imbalance, which motivates resampling techniques such as SMOTE [8]. The proposed framework avoids synthetic resampling and adopts class weighting for compatible models (Section 3.5), since an imbalance close to 85/15 does not warrant introducing interpolated records into a registry whose rows are real administrative aggregates.

Data leakage is a more consequential concern. Systematic review has traced overestimated performance across hundreds of studies in several disciplines to a small set of recurring patterns, chief among them computing preprocessing statistics over the full data set before partitioning [9]. Temporal validation and restriction of all preprocessing to the training partition are the standard remedies [9, 10], and leakage-free evaluation protocols have been proposed as default practice [11]. This framework adopts them (Section efsec:methods:control), and extends them in a direction the leakage literature treats less often. An administrative data set carries fields that are perfectly legitimate descriptors of a closed record yet unavailable at prediction time — publication dates, file-provenance identifiers, and any quantity computed from the outcome year itself. A model using them validates cleanly under a temporal split and remains undeployable. Making that distinction explicit, variable by variable, is one contribution of this paper (Section 3.2).

2.4 Calibration, Decision Utility, and Responsible AI in Judicial Contexts

Discrimination and calibration are distinct properties: a model may rank well while assigning probabilities that are systematically wrong. Calibration has been characterised as the neglected component of predictive analytics [12], and modern high-capacity classifiers are known to be poorly calibrated by default [13, 14]. Where a predicted probability is compared against an action threshold, decision curve analysis [15] evaluates whether acting on the model outperforms the two trivial policies of acting on every unit and acting on none, across the range of thresholds a decision maker might hold. Current reporting frameworks ask explicitly that discrimination, calibration and utility be presented together [16, 17].

Interpretability is a further requirement for institutional adoption in legal settings, where explanations must be intelligible to stakeholders who are not machine learning practitioners [18, 19]. Attribution methods such as SHAP provide local and global explanations of model behaviour [20]. The cautionary evidence is equally pertinent: commercial risk instruments built on more than one hundred variables have been shown to match neither the accuracy nor the fairness advantage their complexity suggests [21], which is why predictive accuracy alone cannot justify deployment in a justice setting [22]. The framework incorporates all three assessments and, where they conflict, allows the most conservative to govern the conclusion.

3 Materials and Methods

Figure 1 summarises the seven stages of CTVEMF.

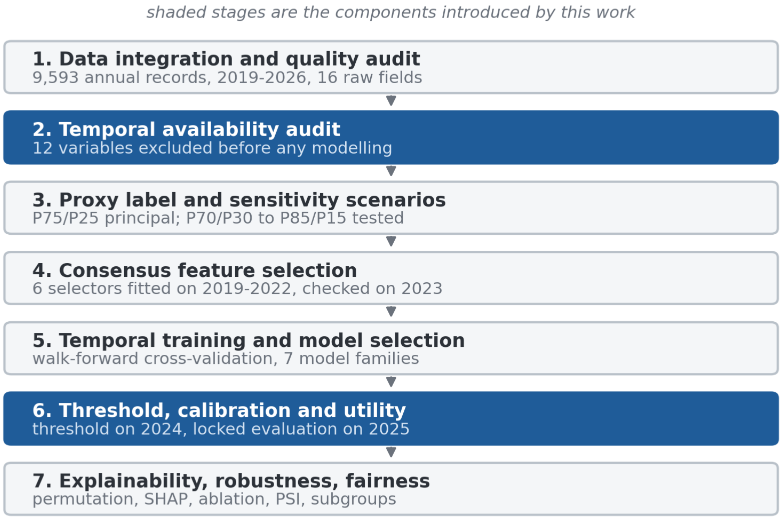

Fig. 1 Stages of the CTVEMF framework. The temporal availability audit (stage 2) and the joint threshold, calibration and utility assessment (stage 6) are the components introduced by this work relative to the conventional predictive pipeline.

3.1 Dataset

3.1.1 Description

This study uses the administrative records of the "MPFN Fiscal Cases" data set, published by the Public Prosecutor's Office — Office of the Attorney General of Peru (MPFN) on the National Open Data Platform [23]. The registry reports the volume of cases received and processed nationwide, disaggregated by prosecutorial district, prosecution office type, subject matter, specialty and case type, distributed in annual CSV files covering 2019–2026.

The unit of analysis is one *operational unit-year*: the annual aggregate for a specific combination of territory, office type, subject matter, specialty and case type. Table 1 reports the composition of the integrated file.

3.1.2 Variables

The official MPFN data dictionary comprises 16 raw administrative fields structured into four dimensions: temporality, institutional geography, case taxonomy and volumetric workload. These fields are the basis of the feature engineering described in Section 3.3.4 and of the availability audit of Section 3.2.

Table 1 Composition of the integrated data set and prevalence of the proxy label by year

Year	Records	Cases received	Cases processed	Proxy risk (%)	Coverage
2019	1,196	1,400,736	1,220,349	11.54	Full year
2020	1,142	863,288	688,925	19.44	Full year
2021	1,252	1,298,697	1,140,369	13.42	Full year
2022	1,272	1,404,619	1,275,920	12.42	Full year
2023	1,214	1,503,919	1,376,760	13.51	Full year
2024	1,195	1,527,838	1,406,901	13.31	Full year
2025	1,199	1,492,353	1,363,570	14.35	Full year
2026	1,123	622,744	530,831	20.21	January–May
Total	9,593	10,114,194	9,003,625	14.68	

Proxy risk is the prevalence of the P75/P25 label of Section 3.3.2. The 2026 row covers five months and its totals are not comparable with the preceding rows.

Two fields carry missing values. The specialised-unit descriptor is absent in 72.39% of records (6,944 rows), which reflects that most units are not designated as specialised; it is encoded as an explicit category rather than imputed. The processed-case count is absent in 0.27% (26 rows) and is imputed with the training-partition median. A binary missingness indicator is retained for each affected field.

3.1.3 Study Period

The analysis covers 2019 through 2026, providing eight years of temporal data. This span permits year-block validation with seven consecutive walk-forward folds. The 2026 file is partial, covering January to May, and is treated as exploratory throughout.

3.2 Prediction Time Point and Temporal Availability Audit

The framework predicts, for a given operational unit and calendar year Y, whether that unit will exhibit the congestion signal defined in Section 3.3.2 once the records for Y are closed. The prediction is issued *at the start of* Y, before any of the case flow of Y has been observed. Every subsequent design decision follows from this definition.

Each candidate variable was assigned to one of three availability classes, reported in Table 2. *Ex ante* variables are observable before year Y begins: the structural attributes of the operational unit, their second-order interactions, the frequency encodings derived from them on the training partition, calendar indicators fixed in advance, and lagged aggregates computed strictly from years earlier than Y. *Concurrent* variables are observable only as Y unfolds: cases received, cases processed and the three quantities derived from them. *Ex post* variables are observable only once the annual file has been compiled and released: the reporting-period descriptor, the download and cut-off dates, and the file-provenance identifier.

Only ex ante variables are admissible. The concurrent block is doubly inadmissible because the label is constructed from it; the ex post block is inadmissible because such fields are artefacts of publication rather than properties of the prosecutorial process. Twelve variables were removed on these grounds before any model was fitted: the five concurrent volumetric quantities, the five label variants derived from them, one constant file-provenance field and one calendar index collinear with the reporting year.

The audit also exposes a residual issue that the reported configuration does not correct, and which is stated here rather than left implicit. Three one-hot indicators derived from the ex post block — one file-provenance identifier, one download date and one cut-off date — were nominated by the consensus selection of Section 3.3.6 and remain among the 74 features of the reported model. Each fires only for records belonging to a single training year, so they behave as year fixed effects. They cannot inflate the reported evaluation figures, because all three are identically zero for every record in 2024, 2025 and 2026; they do, however, allow the model to absorb year-specific variation during training, which a deployed model could not do. Re-fitting the selection and the model on the ex ante subset alone is therefore a prerequisite for deployment, and is listed as such in Section 6.

Table 2 Temporal availability audit of the candidate predictor blocks

Variable block	Availability	Disposition and rationale
Territorial and institutional attributes; pairwise interactions; training-fitted frequency encodings	Ex ante	**Retained.** Structural properties of the operational unit, stable between years and known before the reporting year opens.
Lagged annual aggregates and year-on-year growth rates by fiscal district	Ex ante	**Retained.** Computed exclusively from years strictly earlier than the target year; no record enters its own history.
Pandemic-period and post-pandemic calendar indicators	Ex ante	**Retained.** Fixed by the calendar before the year begins.
Cases received; cases processed	Concurrent	**Excluded.** Accumulate during the target year and are unknown when the prediction is issued.
Case balance; processing rate; balance ratio	Concurrent	**Excluded.** Derived from the concurrent block and used to construct the label.
Proxy-label variants for the four scenarios; the principal label	Concurrent	**Excluded.** Direct functions of the outcome.
Centred year index	Ex ante	**Excluded.** Collinear with the calendar year and extrapolates beyond the training range.
Reporting-period descriptor; download date; cut-off date; file-provenance identifier	Ex post	**Excluded as a block, with a residual exposure.** Generated when the annual file is compiled and released, and effectively encode the reporting year. Three derived indicators nonetheless survived the consensus selection into the reported model (Section 3.2).

3.3 Methodological Framework

3.3.1 Data Quality

All records underwent a quality assessment covering missing values, outliers and internal inconsistencies before any modelling. Categorical fields were normalised and trimmed, numeric fields coerced with explicit missingness flags, and exact duplicates flagged rather than deleted, so that traceability to the published registry is preserved.

3.3.2 Proxy Target

Peru publishes no official certification of prosecutorial congestion, so the target is an auditable proxy rather than a ground truth. For each operational unit-year we define the case balance $b = R - P$ and the processing rate $r = P/R$, where R and P denote cases received and processed. A unit-year is labelled positive when it lies simultaneously in the upper tail of the balance distribution and the lower tail of the processing-rate distribution:

$$y = \mathbb{I}\big[\, b \geq q_\alpha(b) \;\wedge\; r \leq q_{1-\alpha'}(r) \,\big], \tag{1}$$

where both quantiles are estimated *on the training partition only* and applied unchanged to every later year, so that the definition of the label is fixed before any evaluation data are seen.

3.3.3 Target Sensitivity Analysis

Four threshold scenarios were evaluated (Table 3). P75/P25 was adopted as the principal specification: it yields a training threshold of 11 pending cases and a processing rate of 0.9162, a pooled prevalence of 14.68%, and an acceptable balance between institutional severity and event count. The stricter scenarios are more severe but less stable across years, and the more permissive P70/P30 scenario is retained as an early-warning alternative. The choice was deliberately not made on the basis of the F1-score attained under each scenario, since selecting a label definition by downstream performance would itself introduce optimistic bias.

Table 3 Proxy label scenarios and the temporal stability of their prevalence

	Training thresholds		Prevalence				
Scenario	Balance	Rate	Train	2024	2025	CV	Role
P70/P30	7	0.9352	0.1935	0.1983	0.2002	0.165	Early warning
P75/P25	**11**	**0.9162**	**0.1399**	**0.1331**	**0.1435**	**0.219**	**Principal**
P80/P20	17	0.8889	0.0927	0.0879	0.0901	0.278	Severe overload
P85/P15	29	0.8571	0.0533	0.0544	0.0500	0.379	Extreme cases

CV is the coefficient of variation of annual prevalence across 2019–2026. Thresholds are quantiles of the training partition (2019–2023), held fixed for all later years.

The construct was validated empirically rather than by expert elicitation. Cliff's delta between the positive and negative classes is -0.904 for the case balance and 0.809 for the processing rate, confirming that the label separates the two constructs from which it is built. The association of the label with the administrative attributes that remain admissible as predictors is by contrast weak: Cramér's V reaches at most 0.267 for case type and 0.229 for office type. This asymmetry states the difficulty of the task precisely — the label is defined by quantities the model may not see, and must be recovered from structural attributes only weakly associated with it.

3.3.4 Feature Engineering

Fourteen derived variables were constructed in four families: operational pressure indicators; calendar descriptors; second-order categorical interactions (subject matter × office type, judicial district × case type, office type × specialty); and lagged historical aggregates by fiscal district. The historical family warrants a specific note, since it is the most common source of subtle leakage in panel data of this kind. Each lagged feature is an expanding mean over the *previous* years of the same fiscal district, obtained by shifting the annual series one period before aggregation; values missing at the start of a series are filled with the training-partition median. No record therefore contributes to its own history.

High-cardinality categorical variables were frequency-encoded using frequencies estimated on the training partition alone, with unseen categories mapped to zero. The remaining categorical variables were one-hot encoded, and numerical variables median-imputed and standardised.

3.3.5 Data Leakage Control

All preprocessing operations — scaling, imputation, encoding and the multicollinearity audit — were computed exclusively on the training partition and applied to later partitions with the learned parameters, yielding 564 preprocessed candidate features. Combined with the availability audit of Section 3.2, this addresses both forms of leakage relevant here: the statistical form, in which preprocessing borrows information from future partitions, and the temporal form, in which a predictor is unavailable at the moment the prediction must be issued.

3.3.6 Feature Selection

The consensus strategy combined six complementary methods: mutual information, ANOVA F-test and χ^2 as filters, Random Forest importance as an embedded method, and Boruta [7] and recursive feature elimination with cross-validation as wrappers. Each selector nominated its 40 highest-ranked features, and a stability score was computed as the proportion of selectors nominating each feature; those receiving at least two of six votes were retained, giving a final set of 74.

The selection stage was confined to the earliest part of the training block: selectors and their preprocessing were fitted on 2019–2022 and the resulting subset checked on 2023, a year internal to the training partition and never used for reporting. That internal check gave an F1-score of 0.456, a ROC-AUC of 0.800 and a balanced accuracy of 0.751, indicating that reducing 564 features to 74 did not degrade the available signal.

3.4 Feature Space Validation

Unsupervised validation was conducted on the same leakage-free feature space. A k-means solution with two clusters was selected (silhouette 0.371; bootstrap adjusted Rand index 0.798 ± 0.404), and the resulting partition separates operational profiles by risk without any access to the label. Dimensionality reduction by principal component analysis, t-SNE and UMAP was used as supporting diagnostics. These analyses assess whether territorial and institutional structure emerges from the data without supervision, providing convergent evidence for the proxy construct.

3.5 Experimental Design

3.5.1 Temporal Data Partitioning

Each partition carries exactly one role, fixed before any model was fitted (Figure 2 and Table 4):

1. **Feature selection** — 2019–2022, with an internal check on 2023. Never used for reporting.
2. **Model training and hyperparameter search** — 2019–2023 (n = 6,076), under year-block walk-forward cross-validation internal to this block, so that every inner fold trains on earlier years and validates on a later one.
3. **Decision-threshold selection and conformal calibration** — 2024 (n = 1,195). This is the only partition consulted when choosing the operating point, and it serves no other purpose.
4. **Locked evaluation** — 2025 (n = 1,199). Consulted once, after model and threshold were frozen. Every headline figure in Section 4 refers to this partition.
5. **Exploratory external check** — 2026 (n = 1,123), a five-month file reported separately and never used to support a performance claim.

Because the threshold is selected on 2024 and the evaluation performed on 2025, no quantity reported as a final result was optimised on the data that produced it.

Table 4 Role, size and label prevalence of each partition

Partition	Years	Records	Prevalence (%)	Role
Training	2019–2023	6,076	13.99	Fitting all preprocessors, selectors and models; internal walk-forward CV
Validation	2024	1,195	13.31	Decision-threshold selection and conformal calibration only
Test	2025	1,199	14.35	Single locked evaluation
External	2026	1,123	20.21	Exploratory check; partial year

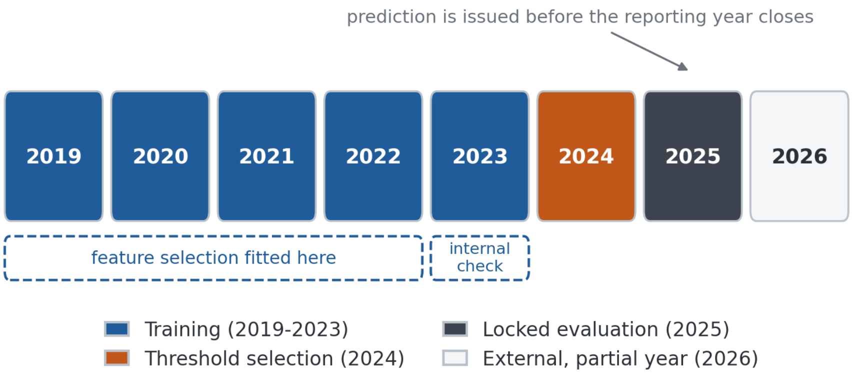

Fig. 2 Temporal partition protocol. Feature selection is confined to 2019–2022 with an internal check on 2023; the decision threshold is chosen on 2024; 2025 is the single locked evaluation set; the partial 2026 file is exploratory only.

3.5.2 Model Configurations

Seven model families were trained on the 74 selected features: a majority-class baseline, ℓ_1/ℓ_2-regularised logistic regression, a calibrated linear support vector machine, XGBoost [24], LightGBM [25], CatBoost [26], and a LightGBM variant tuned with Optuna [27]. Soft-voting and temporally stacked ensembles were also evaluated. Hyperparameters were searched by randomised search under the same year-block walk-forward scheme used for training. All modelling used scikit-learn [28]; Table 5 lists the full stack.

Two benchmark classes frame the comparison. *Single-indicator rules* threshold the case balance or the processing rate directly at their training quantiles; they are reported because they establish what a decision maker who already knows the outcome year's volumes could achieve. An *oracle check* reproduces the conjunction of Equation 1; it attains perfect scores by construction and is reported only as a sanity check, never as a comparator.

Table 5 Software environment

Component	Version	Component	Version	Component	Version
Python	3.12.13	scikit-learn	1.6.1	CatBoost	1.2.10
pandas	2.2.2	LightGBM	4.6.0	SHAP	0.52.0
NumPy	2.0.2	XGBoost	3.3.0	Optuna	4.9.0

3.5.3 Optimization Strategy

Class weighting was applied to handle the observed imbalance (85%/15%) without synthetic resampling. Threshold optimisation was conducted separately from model fitting and exclusively on the 2024 partition of Section efsec:methods:partition, under four criteria reported in Section 4.4 so that the sensitivity of every threshold-dependent metric to that choice is visible.

3.5.4 Evaluation Protocol

Evaluation reports discrimination (ROC-AUC, PR-AUC), threshold-dependent operating characteristics (precision, recall, F1, specificity, Matthews correlation coefficient), calibration (Brier score [29], expected and maximum calibration error, and reliability bins with signed deviations), and decision-analytic utility. Uncertainty is quantified by bootstrap resampling with 300 iterations. Model comparisons use DeLong's test for correlated ROC curves [30], McNemar's test for paired errors [31, 32], and the Friedman test with Nemenyi post-hoc comparison across temporal folds [33].

For decision-analytic utility we compute the net benefit [15]

$$\mathrm{NB}(p_t) = \frac{\mathrm{TP}(p_t)}{n} - \frac{\mathrm{FP}(p_t)}{n} \cdot \frac{p_t}{1-p_t}, \tag{2}$$

where p_t is the threshold probability at which a decision maker is indifferent between acting and not acting, and compare it against flagging every unit ($\mathrm{NB} = \pi - (1-\pi)\, p_t/(1-p_t)$ for prevalence π) and flagging none ($\mathrm{NB} = 0$).

Population stability indices, an ablation study over feature blocks, a stress test under perturbation of the operational attributes, subgroup performance analysis [34] and split-conformal prediction [35, 36] complete the assessment.

4 Results

4.1 Temporal Behaviour of the Workload and the Label

Figure 3 shows annual volumes and label prevalence. Case flow contracted sharply in 2020, with received cases falling 38% relative to 2019, and recovered above pre-pandemic levels by 2022. Label prevalence tracks that disruption, peaking at 19.44% in 2020, settling within 12.4–14.4% from 2021 to 2025, and rising to 20.21% in the partial 2026 file. The 2026 value should not be read as a trend: a five-month window truncates the processing of cases received late in the window and mechanically inflates the measured balance.

4.2 Comparison of Base Models and Ensembles

Table 6 reports every model on the locked 2025 evaluation at the reference threshold of 0.50. Gradient boosting methods dominate the learned models. CatBoost attains the highest ROC-AUC (0.819), while XGBoost and LightGBM are statistically indistinguishable from each other (DeLong $z = -0.41$, $p = 0.68$). Differences between CatBoost and each of the other boosting configurations are statistically significant but numerically small (DeLong $p = 0.0026$–0.0038; absolute AUC differences of 0.023–0.036). The Friedman test across four temporal folds rejects equality of F1 among the five base learners ($\chi^2 = 11.4$, $p = 0.022$), yet the Nemenyi post-hoc comparison identifies no significant pair at the 5% level, the smallest adjusted p-value being 0.056.

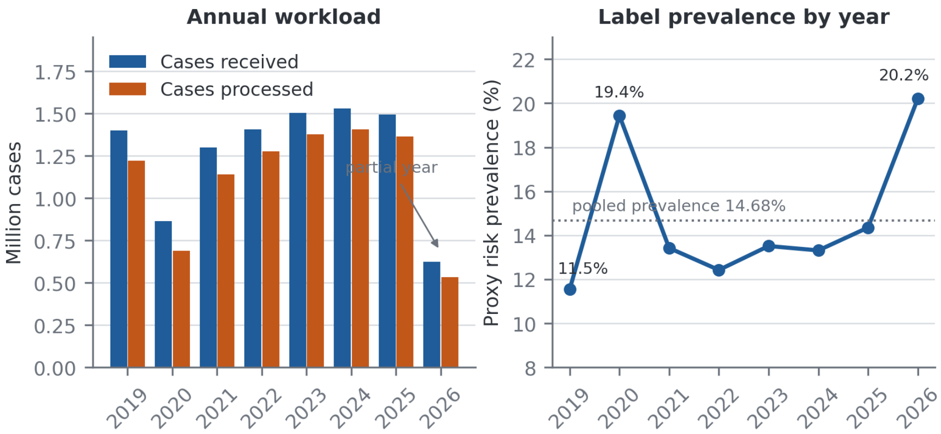

Fig. 3 Annual case volumes (left) and prevalence of the proxy congestion label (right), 2019–2026. The 2026 file covers January to May only.

We therefore report the boosting family as a performance plateau rather than claiming a single winner, and retain XGBoost as the reported configuration for its stable behaviour across the walk-forward folds, where it attains the highest cross-validated F1 (0.559).

Table 6 Performance on the locked 2025 evaluation set at the reference threshold of 0.50

Model	Precision	Recall	F1	ROC-AUC	PR-AUC	MCC
Learned models (admissible predictors only)						
Majority-class baseline	0.000	0.000	0.000	0.500	0.143	0.000
Logistic regression	0.326	0.645	0.433	0.783	0.359	0.327
Support vector machine	0.457	0.122	0.193	0.791	0.370	0.178
XGBoost	0.522	0.407	0.458	0.796	0.400	0.383
LightGBM	0.455	0.471	0.463	0.793	0.502	0.371
CatBoost	0.449	0.541	0.491	**0.819**	0.457	0.399
Soft-voting ensemble	0.488	0.465	0.476	0.817	0.502	0.391
Temporal stacking ensemble	0.368	0.640	0.467	0.817	0.506	0.369
Reference benchmarks (use concurrent information; not deployable)						
Balance-only rule	0.513	1.000	0.679	0.945	0.658	0.657
Processing-rate-only rule	0.548	1.000	0.708	0.914	0.452	0.687
Label-definition oracle	1.000	1.000	1.000	1.000	1.000	1.000

The reference benchmarks are computed from the concurrent variables that the availability audit excludes from the learned models; they bound the problem and are not competitors. The oracle reproduces Equation 1 and is a sanity check only.

The benchmark block is the most informative group in the table. The two single-indicator rules outperform every learned model by a wide margin — an F1 of 0.708 for the processing-rate rule against 0.458 for XGBoost. This does not indicate a poorly constructed pipeline. It measures how much of the label is carried by information that does not exist at the prediction time point: the rules read the outcome year's volumes, and the models are forbidden from doing so. The gap between the two is the price of a genuinely ex ante prediction, and any study reporting rule-level performance from a model trained on concurrent variables is measuring the label's own definition rather than a forecast.

4.3 Temporal Robustness and Uncertainty Quantification

Walk-forward validation (Figure 4) trains on all years up to $t-1$ and evaluates on year t. ROC-AUC remains within 0.834–0.895 across all seven folds with no downward trend, and F1 within 0.542–0.664. Nested temporal cross-validation over four outer folds gives an out-of-fold F1 of 0.549 ± 0.048 for LightGBM, 0.531 ± 0.032 for CatBoost and 0.454 ± 0.040 for logistic regression, reproducing the ordering of Table 6 with fold-to-fold variability that exceeds the between-model differences.

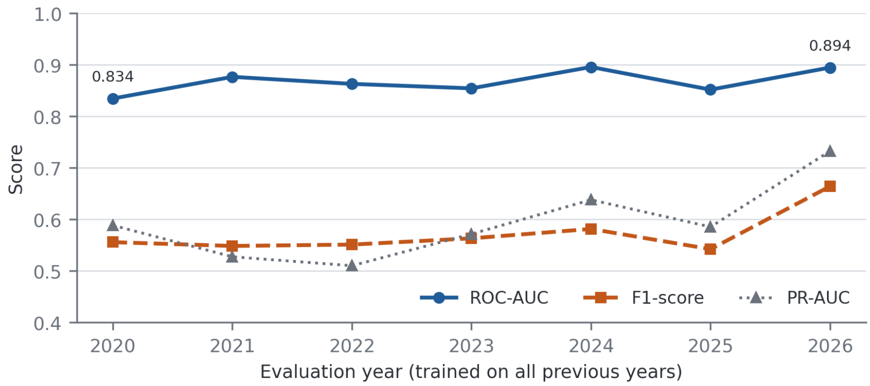

Fig. 4 Walk-forward validation. Each point trains on all preceding years and evaluates on the year shown; discrimination is stable across the seven folds.

The learning curve (Figure 5) shows training F1 declining from 0.948 to 0.900 as the sample grows, while cross-validated F1 rises from 0.217 to 0.490 and then flattens. The residual gap of 0.41 at the largest training size does not close, indicating that the ceiling is set by the weak association between admissible predictors and the label rather than by the volume of data. Adding further years of the same registry would not be expected to alter this.

Population stability indices against the training distribution show no relevant drift for 2024 and 2025 (all PSI < 0.03). For the partial 2026 file, two of five monitored quantities exceed the moderate-drift threshold (processing rate, PSI $= 0.129$; balance ratio, PSI $= 0.138$), consistent with the truncation artefact already noted. Under a stress test perturbing the operational attributes, F1 degrades from 0.409 to 0.279 at a 10% perturbation and to 0.208 at 30%, while ROC-AUC falls only from 0.796 to 0.726. The ranking produced by the model is thus markedly more robust than any fixed-threshold decision derived from it, a finding that anticipates the calibration and utility results below.

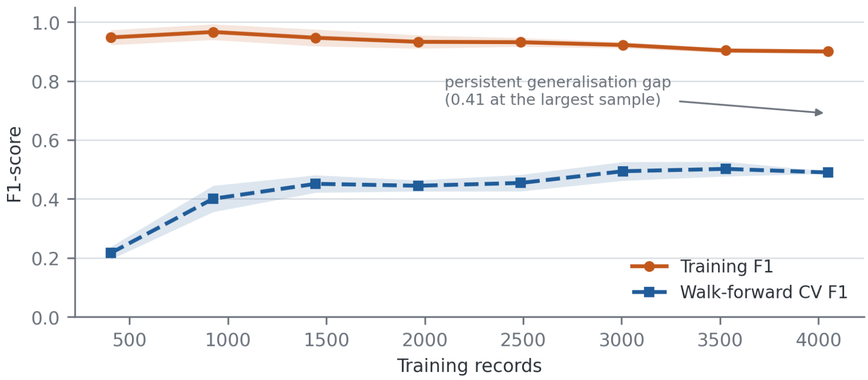

Fig. 5 Learning curve. The persistent gap between training and walk-forward cross-validated F1 reflects the limited signal available in ex ante predictors, not an insufficient sample.

Split-conformal prediction [35] calibrated on the 2024 partition at a nominal coverage of 90% achieves 87.3% empirical coverage on 2025 and 83.7% on the partial 2026 file, with 97.1% and 95.6% singleton prediction sets respectively. A Mondrian variant conditioned on the operational cluster gives 87.6% and 84.8%. Coverage falls short of nominal, modestly on 2025 and more clearly on 2026, consistent with the distribution shift detected for that file.

4.4 Decision Threshold Optimization and Final Model Performance

Table 7 lists the operating points obtainable on the 2024 validation partition under four criteria. The F1-optimal threshold is $\tau = 0.65$; Youden's J is maximised at 0.115, a far more permissive cut; a precision of at least 0.80 requires 0.95; and a recall of at least 0.80 requires 0.20. That these values span almost the entire unit interval is itself a result: the choice of operating point dominates every threshold-dependent metric, and reporting a single F1 without stating how its threshold was chosen conveys very little.

Table 7 Decision thresholds under four criteria, all selected on the 2024 validation partition

Criterion	Threshold	Interpretation
Fixed reference	0.500	Conventional default; rarely optimal under class imbalance
F1-optimal	**0.650**	Maximises F1 on 2024; adopted as the reported operating point
Youden's J	0.115	Maximises sensitivity + specificity -1 ($J = 0.662$)
Precision ≥ 0.80	0.950	Lowest threshold reaching the precision target
Recall ≥ 0.80	0.200	Highest threshold still reaching the recall target

Table 8 reports the frozen model at $\tau = 0.65$. On the locked 2025 evaluation it attains a ROC-AUC of 0.796 (95% CI 0.760–0.838), a PR-AUC of 0.400 (0.330–0.480) and an F1-score of 0.409 (0.331–0.487), from 58 true positives, 54 false positives, 114 false negatives and 973 true negatives. Performance on 2024 is markedly better than on 2025 (F1 of 0.614 against 0.409). This is expected, and it is precisely why the threshold-selection partition cannot be used to report performance: the operating point was fitted to that partition's score distribution.

Table 8 Frozen XGBoost model at the F1-optimal threshold $\tau = 0.65$

Partition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	MCC
Validation 2024	0.901	0.639	0.591	0.614	0.896	0.623	0.558
Test 2025	**0.860**	**0.518**	**0.337**	**0.409**	**0.796**	**0.400**	**0.343**
External 2026	0.820	0.593	0.352	0.442	0.841	0.598	0.359
Bootstrap 95% confidence intervals, test 2025 (300 resamples)							
F1	0.409 (0.331–0.487)						
ROC-AUC	0.796 (0.760–0.838)						
PR-AUC	0.400 (0.330–0.480)						

The 2024 row is reported for completeness only. Because the threshold was selected on that partition, its metrics are optimistically biased and are not comparable with the 2025 row. The 2026 row refers to a five-month file.

4.5 Calibration: Magnitude and Direction of the Deviation

The model attains a Brier score of 0.116, an expected calibration error of 0.089 and a maximum calibration error of 0.461 on the 2025 evaluation. These aggregates conceal a systematic pattern that only the signed deviations reveal (Table 9 and Figure 6).

Table 9 Reliability bins on the 2025 evaluation set, with signed deviations

Predicted band	n	Mean predicted	Observed rate	Predicted − observed	Direction
0.0–0.1	932	0.011	0.072	−0.061	Under-estimation
0.1–0.2	64	0.144	0.234	−0.090	Under-estimation
0.2–0.3	32	0.248	0.219	+0.030	Over-estimation
0.3–0.4	16	0.341	0.375	−0.034	Under-estimation
0.4–0.5	21	0.450	0.333	+0.117	Over-estimation
0.5–0.6	17	0.555	0.529	+0.025	Over-estimation
0.6–0.7	18	0.663	0.444	+0.218	Over-estimation
0.7–0.8	10	0.761	0.300	+**0.461**	Over-estimation
0.8–0.9	34	0.854	0.618	+0.236	Over-estimation
0.9–1.0	55	0.949	0.527	+0.422	Over-estimation

Brier score 0.116; expected calibration error 0.089; maximum calibration error 0.461, attained in the 0.7–0.8 band.

The deviation is directional. In the two lowest bands the model *under-estimates* risk: units assigned a mean probability of 0.011 exhibit an observed rate of 0.072. In every band above 0.3, with the single exception of the sparsely populated 0.3–0.4 band, the model *over-estimates* risk, and the over-estimation grows with confidence.

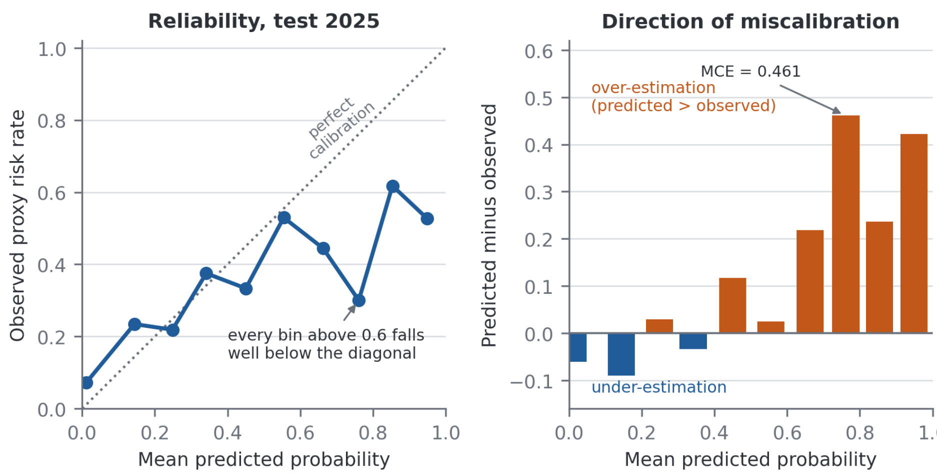

Fig. 6 Calibration on the 2025 evaluation set. Left: observed rate against mean predicted probability. Right: signed deviation by band. Above a predicted probability of 0.3 the model over-estimates risk, severely so in the highest bands.

Units assigned a mean probability of 0.949 exhibit an observed rate of 0.527, and the maximum calibration error of 0.461 arises in the 0.7–0.8 band, where a mean predicted probability of 0.761 corresponds to an observed rate of 0.300.

The direction has a concrete cause and a concrete consequence. The model was trained with cost-sensitive class weighting, which inflates predicted probabilities for the minority class by construction, and the inflation is largest exactly where the model is most confident. Operationally, a raw predicted probability from this model must not be read as an expected frequency of congestion. It ranks units correctly, as Section 4.6 shows, but its absolute scale is not trustworthy. Any deployment must therefore recalibrate the scores, and the recalibration must be fitted on a partition held apart from the final evaluation; the 2024 partition is the natural candidate, with the caveat that it already serves as the threshold-selection set, so using it for both purposes would leave the interaction between them unmeasured. We report this as an outstanding requirement rather than a completed step: the results presented here are those of the uncalibrated model, and no recalibrated performance is claimed.

4.6 Risk Stratification

Grouping the predicted probabilities into four bands (Table 10 and Figure 7) shows that the ranking induced by the model is sound even where its probability scale is not. The observed congestion rate rises monotonically from 7.2% in the lowest band to 52.1% in the highest, against a base rate of 14.35%. The two upper bands jointly contain 14.3% of the operational units and 48.3% of the congestion events, while the lowest band holds 77.7% of the units at less than half the base rate.

Two qualifications are essential. First, the boundaries lie on the uncalibrated scale characterised in Section 4.5: the mean predicted probability in the highest band is 0.807 against an observed rate of 0.521, so the band labels describe rank position rather than

Table 10 Risk stratification on the 2025 evaluation set

Stratum	Band	n	Events	Observed rate	Share of units	Share of events
Low	< 0.10	932	67	0.072	77.7%	39.0%
Moderate	0.10–0.30	96	22	0.229	8.0%	12.8%
High	0.30–0.60	54	22	0.407	4.5%	12.8%
Very high	≥ 0.60	117	61	0.521	9.8%	35.5%
All		1,199	172	0.144	100%	100%

Bands are deciles of predicted probability aggregated into four categories. Because the boundaries are defined on the uncalibrated probability scale, they are not transferable to a recalibrated model and would require re-derivation and independent validation before use.

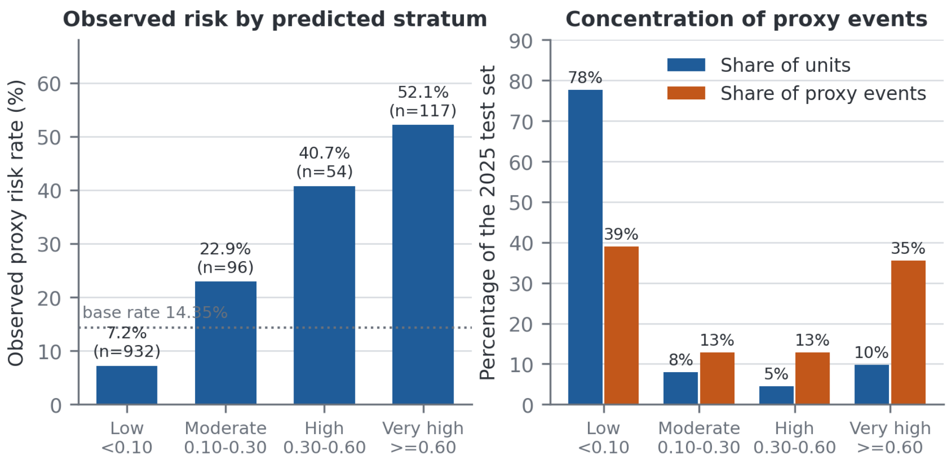

Fig. 7 Risk stratification on the 2025 evaluation set. Left: observed congestion rate by predicted stratum, against the base rate. Right: the share of operational units and of congestion events in each stratum.

risk magnitude. Second, the strata are derived from the same 2025 partition on which they are described. An independent validation of the boundaries on data not used to construct them has not been performed, and until it is, the stratification should be read as a description of the model's ranking behaviour and not as a validated triage instrument.

4.7 Decision-Analytic Utility

Discrimination and stratification say nothing about whether acting on the model beats the alternatives available to a decision maker. Table 11 and Figure 8 apply Equation 2 across the range of threshold probabilities and compare the model against flagging every unit and flagging none.

The result is mixed, and its negative half is the more important. For threshold probabilities between 0.1 and 0.5 the model carries a positive net benefit and dominates both defaults; at $p_t = 0.15$ its advantage of 0.060 over the better default corresponds

Table 11 Net benefit on the 2025 evaluation set

p_t	Units flagged	TP	FP	Model	Flag all	Preferred strategy
0.10	267	105	162	+0.073	+0.048	Model
0.20	203	90	113	+0.052	−0.071	Model
0.30	171	83	88	+0.038	−0.224	Model
0.40	155	77	78	+0.021	−0.428	Model
0.50	134	70	64	+0.005	−0.713	Model
0.65	**112**	**58**	**54**	**−0.035**	−1.591	**Flag none**
0.70	99	53	46	−0.045	−1.855	Flag none
0.80	89	50	39	−0.088	−3.283	Flag none
0.90	55	29	26	−0.171	−7.566	Flag none

The row in bold is the F1-optimal operating point adopted in Table 8. Flagging no unit yields a net benefit of zero by definition.

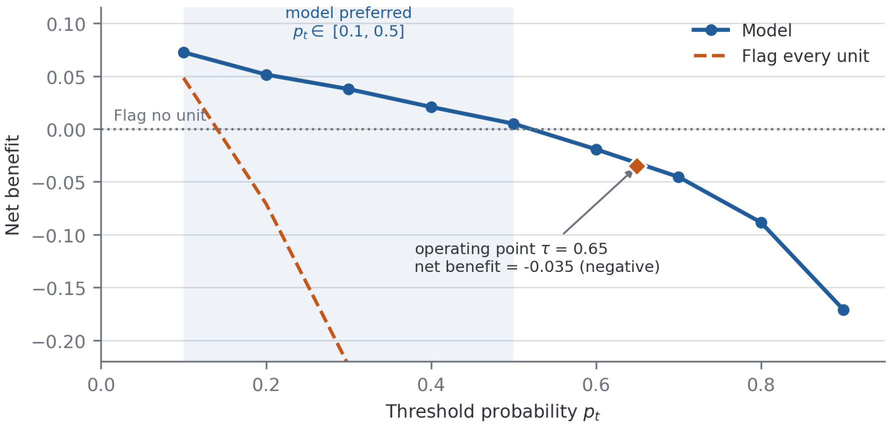

Fig. 8 Decision curve analysis on the 2025 evaluation set. The model exceeds both default strategies for threshold probabilities between 0.1 and 0.5. At the F1-optimal operating point $\tau = 0.65$ its net benefit is negative, meaning that flagging no unit would be preferable for a decision maker holding that threshold.

to correctly identifying roughly six additional congested units per hundred without increasing the number of unnecessary interventions. But at the F1-optimal operating point of $\tau = 0.65$ — the point at which the headline metrics of Table 8 are computed — the net benefit is -0.035. It is negative, and below that of the flag-none strategy. A decision maker genuinely requiring a 65% probability of congestion before committing resources would be better served by intervening nowhere than by following this model's recommendations.

The conclusion this compels is that the threshold maximising F1 is not a threshold at which the model is operationally useful, and the two criteria are not interchangeable. The framework supports identifying and ranking units for review under moderate intervention costs — that is, for decision makers whose indifference threshold lies between roughly 0.1 and 0.5 — and does not support a high-confidence alerting system. No claim of operational utility is made at the F1-optimal operating point.

4.8 Ablation Study of Feature Groups

Systematic ablation (Figure 9) confirms that territorial and institutional information carries most of the predictive weight, and delivers one counter-intuitive result. Removing the territorial block costs 0.053 F1, the largest degradation of any single block. Removing the frequency encodings or the categorical interactions produces smaller changes. Removing the historical and growth block, however, *improves* F1 by 0.031 and ROC-AUC by 0.026. The lagged aggregates therefore do not contribute to predictive performance on the locked evaluation set despite their prominence in the attribution rankings; they appear to add variance without adding signal. We report this rather than suppress it: on the present evidence the historical block does not earn its place in the model.

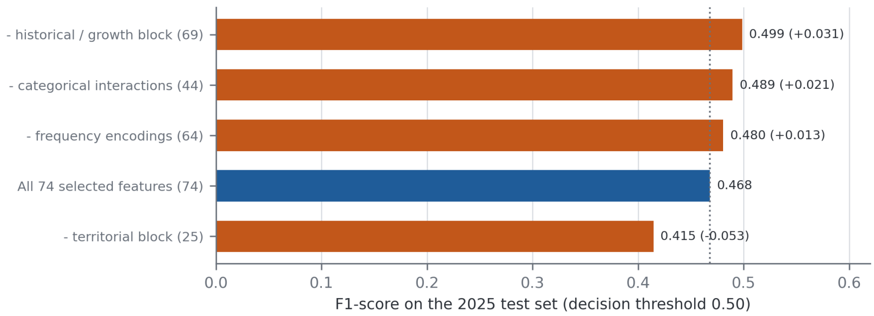

Fig. 9 Ablation study on the 2025 evaluation set at the reference threshold of 0.50. Removing the territorial block degrades performance most; removing the historical and growth block improves it.

4.9 Explainability: Global Feature Importance

Permutation importance computed on the reported XGBoost model and mean absolute SHAP values [20] computed on the tuned LightGBM configuration are shown in Figure 10. Both are reported, each labelled with the estimator it was computed on, because the two attributions were derived from different models and are not interchangeable; their agreement is a property worth reporting rather than an assumption to be made silently.

Both rankings are led by the frequency encoding of the judicial district $\times$ case type interaction (permutation importance 0.121; mean |SHAP| 1.668), followed by the frequency of the specialised-unit designation, the frequency of the province, and the indicator for superior-level prosecution offices. The lagged historical aggregates occupy the mid-range of both rankings. The picture is coherent: what the model can learn from ex ante information is the structural composition of an operational unit — where it sits, what matters it handles, and how common that configuration is in the registry — rather than any dynamic property of its case flow.

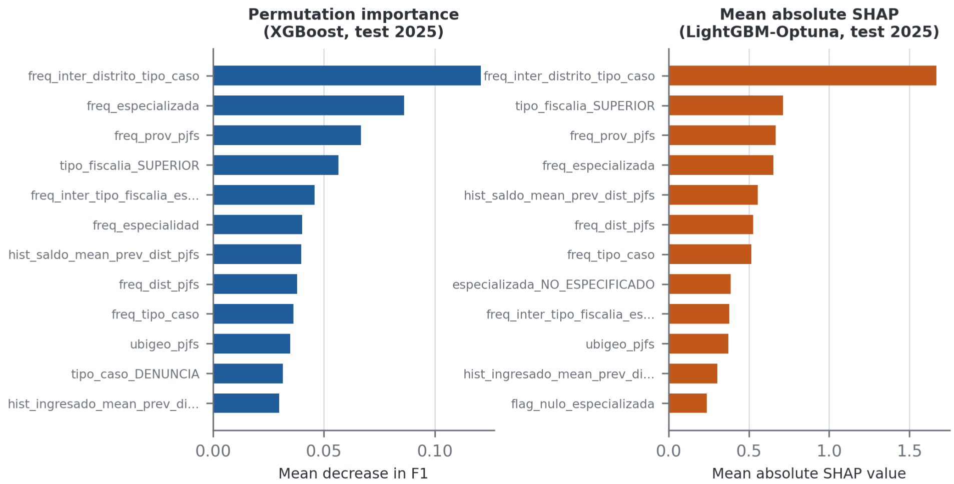

Fig. 10 Global variable importance. Left: permutation importance on the reported XGBoost model. Right: mean absolute SHAP values on the tuned LightGBM configuration. The two attributions were computed on different estimators and are shown separately for that reason.

4.10 Unsupervised Validation and Territorial Fairness

Unsupervised profiling with k-means on the leakage-free feature space selects a two-cluster solution. The smaller cluster (1,269 units) exhibits a congestion rate of 23.9% against 13.3% in the larger one, and this ordering is preserved under all four label scenarios. That an unsupervised partition of the admissible predictors recovers the same risk ordering as the supervised label provides independent, if modest, support for the proxy construct.

Subgroup analysis by territory reveals a substantial disparity. In the Lima metropolitan area (258 units, observed prevalence 16.7%) the model flags 22.1% of units and attains a true positive rate of 0.535 with a false positive rate of 0.158. In the provinces (941 units, observed prevalence 13.7%) it flags only 5.8% of units, with a true positive rate of 0.271 and a false positive rate of 0.025. The equal-opportunity gap of 0.264 means that a congested provincial unit is roughly half as likely to be identified as an equivalently congested metropolitan one. Since the model relies heavily on frequency encodings and metropolitan configurations are more frequent in the registry, this disparity is a predictable consequence of the feature design rather than an incidental artefact, and any institutional use would have to correct for it, for instance by calibrating and thresholding within territorial strata.

5 Interpretability and Impact

5.1 Interpretability

The framework prioritises interpretability through permutation importance and SHAP values, which together provide global explanations that are intelligible to judicial and administrative stakeholders without extensive machine learning expertise. Because the admissible feature set contains only structural attributes and lagged aggregates, each attribution admits a direct institutional reading: a unit is ranked highly because of where it sits, what class of matters it handles and how that configuration has behaved in previous years, never because of information about the year being predicted.

5.2 Operational Impact

Read conservatively, the framework supports a review-prioritisation workflow: rank operational units by predicted score, direct planning attention to the upper strata, and use the conformal prediction sets to identify units on which the model declines to commit. The concentration reported in Table 10 is the operational argument — one tenth of the units captures more than a third of the congestion events.

Three uses are not supported by the present evidence. Automated alerting is excluded by the negative net benefit at the F1-optimal operating point (Section 4.7). Resource allocation proportional to predicted probability is excluded by the systematic over-estimation above 0.3 (Section 4.5), which would over-allocate to the highest-scoring units. Any use as an equitable prioritisation tool is excluded until the territorial disparity is corrected. A monetary impact figure derived from the confusion matrix would depend entirely on the assumed ratio between the cost of a missed congested unit and that of an unnecessary review; the net benefit analysis is the principled way to express that dependence, and it is reported instead.

5.3 Discussion of Underlying Mechanisms

The analysis indicates that the predictable component of prosecutorial congestion risk is structural rather than dynamic: territorial and institutional composition dominates, while lagged case-flow history contributes nothing measurable on the locked evaluation. The gap to the single-indicator rules quantifies the remainder. Those rules reach F1 scores near 0.70 by reading the outcome year's volumes; the learned models reach 0.46 without them. That difference is not a deficiency of the models but a measurement of the intrinsic difficulty of the ex ante problem, and it is visible only because the availability audit forced the two information sets apart. In a study without such an audit the volumetric variables would very likely have entered the feature set, the reported F1 would have approached 0.70, and the resulting figure would have described a model that cannot be run at the moment its prediction is needed.

6 Limitations

The label is a proxy constructed from a quantile rule, not an official certification of congestion. A unit flagged by this framework has a high case balance and a low processing rate, which is a plausible but not authoritative indicator of institutional overload. The proxy was validated empirically and by convergence with an unsupervised partition, but no expert panel was convened, and a formal Delphi or analytic hierarchy process elicitation remains outstanding.

The reported model is not recalibrated. Calibration was measured and characterised but not corrected, and the risk bands of Section 4.6 are consequently defined on an uncalibrated scale and have not been validated on data independent of those used to describe them. Both steps are prerequisites for deployment.

The reported model also retains the three publication-metadata indicators identified in Section 3.2. Because each is identically zero throughout 2024, 2025 and 2026, they cannot have inflated any figure reported here, but their presence during training is inconsistent with the availability criterion the framework sets for itself. Re-executing the consensus selection and the model fit over the ex ante subset alone is the outstanding correction.

The decision curve analysis is computed on the deciles of the predicted-probability distribution of the 2025 evaluation set and reports net benefit at the resulting grid of threshold probabilities; a finer grid would smooth the curve without changing the sign of the net benefit at the reported operating point.

The unit of analysis is an annual aggregate, which bounds the temporal resolution of any prediction to one year and prevents the framework from anticipating within-year dynamics. The registry records nothing on staffing, budget, procedural complexity or case severity, all of which plausibly influence processing capacity; their absence is a substantive constraint on achievable performance, and the persistent learning-curve gap is consistent with it. The 2026 file is partial and its results are exploratory throughout. Finally, causality cannot be inferred from observational data, and the framework has been evaluated on a single national registry, so its transferability to other prosecution services or judicial systems remains an open question.

7 Conclusions

This study proposes and audits CTVEMF, a reproducible machine learning framework for estimating the proxy risk of prosecutorial congestion that integrates consensus-based feature selection, temporal availability auditing, year-block validation, robustness assessment and explainability within a single workflow.

The main contributions are: (i) a consensus feature selection strategy over six complementary methods with explicit multicollinearity auditing, confined to the earliest training block; (ii) a temporal availability audit that removed twelve variables — among them the five defining the label — before any model was fitted, and that additionally exposed three publication-metadata indicators surviving into the reported feature set; (iii) a partition protocol in which threshold selection and final evaluation are strictly disjoint; (iv) systematic robustness assessment through ablation, stress testing, drift monitoring and unsupervised validation; and (v) a joint discrimination–calibration–utility assessment.

On the locked 2025 evaluation the framework attains a ROC-AUC of 0.796 (95% CI 0.760–0.838) and an F1-score of 0.409 (0.331–0.487), with stable discrimination across seven walk-forward folds. The joint assessment also yields two findings that constrain the claims: risk is systematically over-estimated above a predicted probability of 0.3, and the net benefit at the F1-optimal operating point is negative. The framework is therefore reported as a ranking and triage instrument, useful to decision makers whose intervention threshold lies between roughly 0.1 and 0.5, and not as an alerting system. The methodological lesson is that the three audits together change the conclusion:

discrimination alone would have supported a considerably stronger claim than the evidence warrants.

Future work should complete the recalibration and the independent validation of the risk bands, re-fit the selection over the ex ante subset, address the territorial disparity through stratified thresholding, incorporate capacity and complexity variables absent from the current registry, and test the transferability of the framework to other jurisdictions.

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Declarations

Funding No funding was received for this research.
Competing Interests The authors declare that they have no competing interests.
Ethics Approval Not applicable. This study uses publicly available aggregate administrative data released under an open data licence and involves no human subjects.
Consent to Participate Not applicable.
Consent for Publication Not applicable.
Data Availability The "MPFN Fiscal Cases" data set is publicly available from the Plataforma Nacional de Datos Abiertos at https://www.datosabiertos.gob.pe/dataset/mpfn-casos-fiscales. The derived tables and figures reported in this article are included as supplementary material.
Code Availability The complete analysis pipeline, including the availability audit, the partition protocol, the selection and modelling stages and the figure-generation scripts, is available in the repository accompanying this manuscript.
Authors' Contributions M.E.G. conceptualised the study, designed the methodology, implemented the pipeline and drafted the manuscript. J.A.C.L. contributed to the experimental design, the validation of results and the revision of the manuscript. Both authors reviewed and approved the final version.

References

[1] Azaria S, Ronen B, Shamir N. Alleviating Court Congestion: The Case of the Jerusalem District Court. INFORMS Journal on Applied Analytics. 2024 5;54(3):267–281. https://doi.org/10.1287/inte.2023.0026.

[2] Vasconcelos FF, Sátiro RM, Fávero LPL, Bortoloto GT, Corrêa HL. Analysis of Judiciary Expenditure and Productivity Using Machine Learning Techniques. Mathematics. 2023 7;11(14):3195. https://doi.org/10.3390/math11143195.

[3] Pernici B, Bono CA, Piro L, Del Treste M, Vecchi G. Improving the analysis of the judiciary performance – the use of data mining techniques to assess the

timeliness of civil trials. IJPSM. 2024 1;37(1):59–76. https://doi.org/10.1108/IJPSM-02-2023-0058.

[4] Alhalalmeh AH, Al-Tarawneh A. Artificial Intelligence and the Law: The Complexities of Technology and Legalities. In: Hannoon A, Mahmood A, editors. Intelligence-Driven Circular Economy. vol. 1174. Springer Nature Switzerland; 2025. p. 641–649.

[5] Zhou X, Yuan W, Gao Q, Yang C. An efficient ensemble learning method based on multi-objective feature selection. Information Sciences. 2024 9;679:121084. https://doi.org/10.1016/j.ins.2024.121084.

[6] Moslemi A. A tutorial-based survey on feature selection: Recent advancements on feature selection. Engineering Applications of Artificial Intelligence. 2023 11;126:107136. https://doi.org/10.1016/j.engappai.2023.107136.

[7] Kursa MB, Rudnicki WR. Feature Selection with The Boruta Package. Journal of Statistical Software. 2010 1;36(11). https://doi.org/10.18637/jss.v036.i11.

[8] Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP. SMOTE: Synthetic Minority Over-sampling Technique. Journal of Artificial Intelligence Research. 2002 6;16:321–357. https://doi.org/10.1613/jair.953.

[9] Kapoor S, Narayanan A. Leakage and the reproducibility crisis in machine-learning-based science. Patterns. 2023 8;4(9):100804. https://doi.org/10.1016/j.patter.2023.100804.

[10] Sasse L, Nicolaisen-Sobesky E, Dukart J, Eickhoff SB, Götz M, Hamdan S, et al. On Leakage in Machine Learning Pipelines. arXiv preprint arXiv:231104179. 2023;https://doi.org/10.48550/ARXIV.2311.04179.

[11] Hamdan S, More S, Sasse L, Komeyer V, Patil KR, Raimondo F. Julearn: an easy-to-use library for leakage-free evaluation and inspection of ML models. arXiv preprint arXiv:231012568. 2023;https://doi.org/10.48550/ARXIV.2310.12568.

[12] Van Calster B, McLernon DJ, van Smeden M, Wynants L, Steyerberg EW. Calibration: the Achilles heel of predictive analytics. BMC Medicine. 2019;17:230. https://doi.org/10.1186/s12916-019-1466-7.

[13] Guo C, Pleiss G, Sun Y, Weinberger KQ. On calibration of modern neural networks. In: Proceedings of the 34th International Conference on Machine Learning; 2017. p. 1321–1330.

[14] Niculescu-Mizil A, Caruana R. Predicting good probabilities with supervised learning. In: Proceedings of the 22nd International Conference on Machine Learning; 2005. p. 625–632.

[15] Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction models. Medical Decision Making. 2006;26(6):565–574. https://doi.org/10.1177/0272989X06295361.

[16] Collins GS, Moons KGM, Dhiman P, Riley RD, Beam AL, Van Calster B, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ. 2024;385:e078378. https://doi.org/10.1136/bmj-2023-078378.

[17] Steyerberg EW, Vickers AJ, Cook NR, Gerds T, Gonen M, Obuchowski N, et al. Assessing the performance of prediction models: a framework for traditional and novel measures. Epidemiology. 2010;21(1):128–138. https://doi.org/10.1097/EDE.0b013e3181c30fb2.

[18] Richmond KM, Muddamsetty SM, Gammeltoft-Hansen T, Olsen HP, Moeslund TB. Explainable AI and Law: An Evidential Survey. DISO. 2024 5;3(1):1. https://doi.org/10.1007/s44206-023-00081-z.

[19] Saarela M, Podgorelec V. Recent Applications of Explainable AI (XAI): A Systematic Literature Review. Applied Sciences. 2024 10;14(19):8884. https://doi.org/10.3390/app14198884.

[20] Lundberg SM, Lee SI. A Unified Approach to Interpreting Model Predictions. arXiv preprint arXiv:170507874. 2017;https://doi.org/10.48550/arxiv.1705.07874.

[21] Dressel J, Farid H. The accuracy, fairness, and limits of predicting recidivism. Science Advances. 2018 1;4(1):eaao5580. https://doi.org/10.1126/sciadv.aao5580.

[22] Taylor I. Is explainable AI responsible AI? AI & Society. 2025 3;40(3):1695–1704. https://doi.org/10.1007/s00146-024-01939-7.

[23] Ministerio Público – Fiscálía de la Nación (MPFN). [MPFN] Casos fiscales – Plataforma Nacional de Datos Abiertos. Plataforma Nacional de Datos Abiertos. 2022 4;Dataset. Available from: https://www.datosabiertos.gob.pe/dataset/mpfn-casos-fiscales.

[24] Chen T, Guestrin C. XGBoost: a scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining; 2016. p. 785–794.

[25] Ke G, Meng Q, Finley T, Wang T, Chen W, Ma W, et al. LightGBM: a highly efficient gradient boosting decision tree. In: Advances in Neural Information Processing Systems 30; 2017. p. 3146–3154.

[26] Prokhorenkova L, Gusev G, Vorobev A, Dorogush AV, Gulin A. CatBoost: unbiased boosting with categorical features. In: Advances in Neural Information

Processing Systems 31; 2018. p. 6638–6648.

[27] Akiba T, Sano S, Yanase T, Ohta T, Koyama M. Optuna: a next-generation hyperparameter optimization framework. In: Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining; 2019. p. 2623–2631.

[28] Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: machine learning in Python. Journal of Machine Learning Research. 2011;12:2825–2830.

[29] Brier GW. Verification of forecasts expressed in terms of probability. Monthly Weather Review. 1950;78(1):1–3.

[30] DeLong ER, DeLong DM, Clarke-Pearson DL. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. Biometrics. 1988;44(3):837–845. https://doi.org/10.2307/2531595.

[31] McNemar Q. Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika. 1947;12(2):153–157. https://doi.org/10.1007/BF02295996.

[32] Dietterich TG. Approximate statistical tests for comparing supervised classification learning algorithms. Neural Computation. 1998;10(7):1895–1923. https://doi.org/10.1162/089976698300017197.

[33] Demšar J. Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research. 2006;7:1–30.

[34] Hardt M, Price E, Srebro N. Equality of opportunity in supervised learning. In: Advances in Neural Information Processing Systems 29; 2016. p. 3315–3323.

[35] Vovk V, Gammerman A, Shafer G. Algorithmic Learning in a Random World. New York: Springer; 2005.

[36] Angelopoulos AN, Bates S. Conformal prediction: a gentle introduction. Foundations and Trends in Machine Learning. 2023;16(4):494–591. https://doi.org/10.1561/2200000101.